



instructables classes

ELECTRONICS CLASS (/class/Electronics-Class/)

LESSON 1: GETTING STARTED WITH ELECTRONICS

with randofu

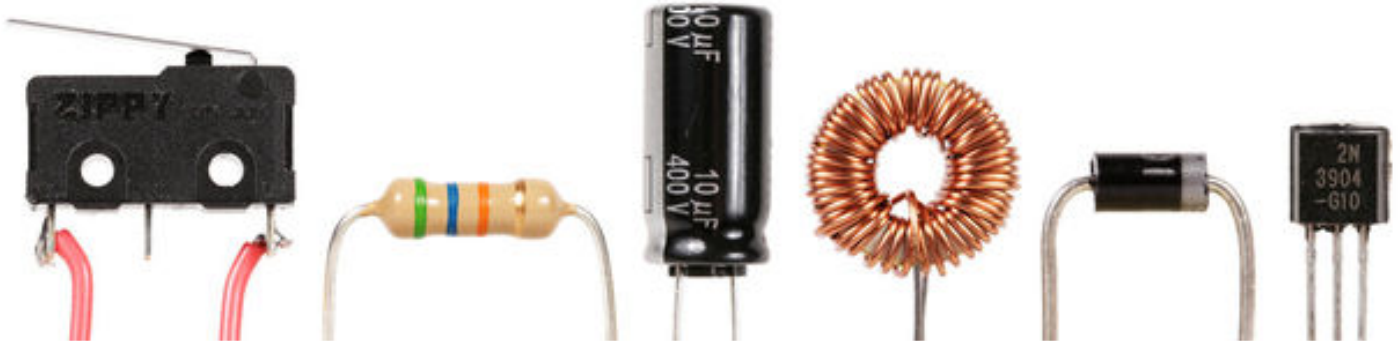
Whether you have a project in mind, or simply wanted to learn but didn't know where to start, getting into electronics is much easier than you may think. You don't even need a degree in electrical engineering to figure this out - I don't! While it may seem foreign and intimidating at first, anyone can do it.

Throughout this class, we are going to cover electronics from the fundamentals of electricity all the way through to microchips. At each stage you will build a project, and begin to grasp these concepts in a hands-on practical way.



What Is Electronics?

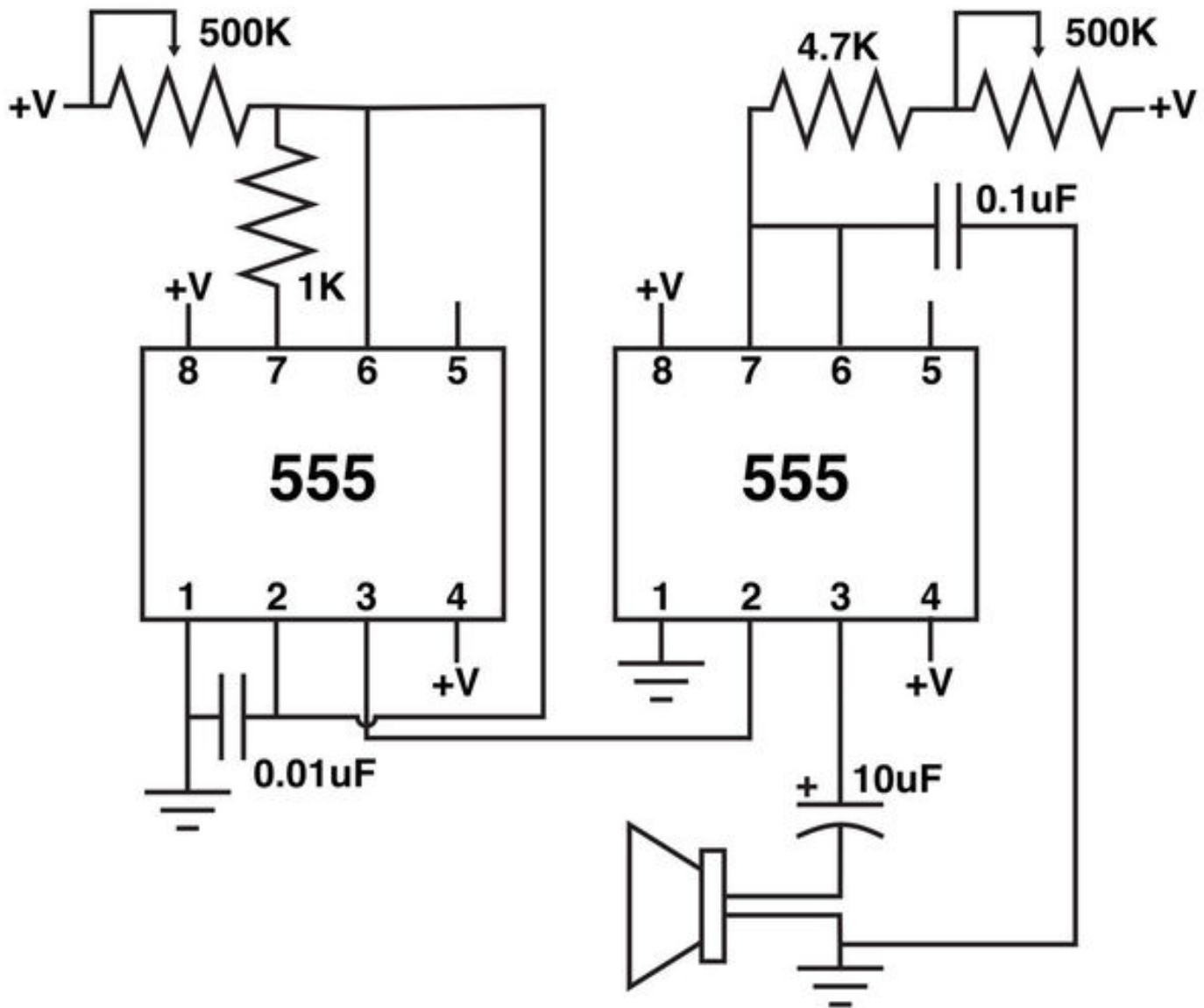
Electronics is the science and technology concerned with regulating flow of electrical current. Put simply, it lets you move electrons around. While this may seem like not that big of a deal, the regulation of electrons has enabled some of the most important innovations of the last century including computers, televisions, rocketships, electric cars, rocketships, video games, smartphones, Tickle-Me-Elmo, hoverboards (both those that really hover and those that do not), and - of course - rocketships.



Schematics and Symbols

Schematics are blueprints for arranging components and constructing electronic circuits. They may appear to be a cryptic mess of indecipherable symbols, but once you get the hang of it, are actually easy to understand. By the end of this class, you will understand how to read this schematic and build this circuit.

Throughout the class as I introduce each new component, I will also include its schematic symbol. This will help you begin to understand what these drawings mean.



Materials and Tools

The art of electronics requires you to procure a lot of materials and tools. Some of this you may already have, but most of it you won't. It's a little bit of an investment and some of it may seem intimidating. Don't worry if it does. It will all make sense to you in due time.

Below is a list of the tools you will need for this class. We will go over each of them in more depth a moment.

- (x1) [Basic electronics kit \(https://www.amazon.com/Elenco-Component-Assorted-Electronic-Components/dp/B0002HBQHC/\)](https://www.amazon.com/Elenco-Component-Assorted-Electronic-Components/dp/B0002HBQHC/)
- (x1) [22AWG stranded core wire \(https://www.amazon.com/Electronix-Express-Hook-Stranded-Gauge/dp/B00B4ZQ3L0/\)](https://www.amazon.com/Electronix-Express-Hook-Stranded-Gauge/dp/B00B4ZQ3L0/)
- (x1) [22AWG solid core wire \(https://www.amazon.com/Electronix-Express--Hook-Wire-Solid/dp/B00B4ZRPEY/\)](https://www.amazon.com/Electronix-Express--Hook-Wire-Solid/dp/B00B4ZRPEY/)
- (x3) [Breadboards \(https://www.amazon.com/Frentaly%C2%AE-Solderless-BreadBoard-tie-points-\)](https://www.amazon.com/Frentaly%C2%AE-Solderless-BreadBoard-tie-points-)

[power/dp/B01258UZMC/](https://www.amazon.com/dp/B01258UZMC/))

(x1) [Multimeter \(https://www.amazon.com/Flexzion-Multimeter-Automotive-Electrical-Components/dp/B017QR594G/\)](https://www.amazon.com/Flexzion-Multimeter-Automotive-Electrical-Components/dp/B017QR594G/)

(x1) [Test amp \(https://www.radioshack.com/products/radioshack-mini-audio-amplifier?variant=5717614213\)](https://www.radioshack.com/products/radioshack-mini-audio-amplifier?variant=5717614213)

(x1) [Wire cutter \(http://www.amazon.com/Tools-VISE-GRIP-Stripper-Crimper-2078309/dp/B000JNNWQ2/?tag=instructabl09-20\)](http://www.amazon.com/Tools-VISE-GRIP-Stripper-Crimper-2078309/dp/B000JNNWQ2/?tag=instructabl09-20)

(x1) [Snippers \(http://www.amazon.com/Hakko-CHP-170-Stand-off-Construction-21-Degree/dp/B017ODDPNO/?tag=instructabl09-20\)](http://www.amazon.com/Hakko-CHP-170-Stand-off-Construction-21-Degree/dp/B017ODDPNO/?tag=instructabl09-20)

(x1) [Heat gun \(http://www.amazon.com/Kawasaki-840015-Black-10-Piece-Heat/dp/B000H4I67I/?tag=instructabl09-20\)](http://www.amazon.com/Kawasaki-840015-Black-10-Piece-Heat/dp/B000H4I67I/?tag=instructabl09-20)

(x1) [Shrink tube \(http://www.amazon.com/SummitLink-Assorted-Shrink-Colors-Tubing/dp/B00LVFDLUO/?tag=instructabl09-20\)](http://www.amazon.com/SummitLink-Assorted-Shrink-Colors-Tubing/dp/B00LVFDLUO/?tag=instructabl09-20)

(x1) [Jumper cables \(http://www.amazon.com/SE-TL10-10-Piece-Alligator-Clips/dp/B0002KRABU/?tag=instructabl09-20\)](http://www.amazon.com/SE-TL10-10-Piece-Alligator-Clips/dp/B0002KRABU/?tag=instructabl09-20)

(x1) [Battery holders \(https://www.amazon.com/3Pcs-Battery-Holder-Standard-Connector/dp/B014YSFMXI/\)](https://www.amazon.com/3Pcs-Battery-Holder-Standard-Connector/dp/B014YSFMXI/)

(x1) [Power drill \(http://www.amazon.com/DEWALT-DC970K-2-18-Volt-Compact-Driver/dp/B002RLR0EY/?tag=instructabl09-20\)](http://www.amazon.com/DEWALT-DC970K-2-18-Volt-Compact-Driver/dp/B002RLR0EY/?tag=instructabl09-20)

(x1) [Drill bits \(http://www.amazon.com/DEWALT-DW1354-14-Piece-Titanium-Drill/dp/B0045PQ762/?tag=instructabl09-20\)](http://www.amazon.com/DEWALT-DW1354-14-Piece-Titanium-Drill/dp/B0045PQ762/?tag=instructabl09-20)

(x1) [Screwdrivers \(http://www.amazon.com/Craftsman-Screwdriver-Phillips-Slotted-Made/dp/B006YVTAFU/?tag=instructabl09-20\)](http://www.amazon.com/Craftsman-Screwdriver-Phillips-Slotted-Made/dp/B006YVTAFU/?tag=instructabl09-20)

(x1) [Mini screwdriver set \(http://www.amazon.com/Herco%C2%AE-HE826-Precision-Screwdriver-Set/dp/B000CCUFT2/?tag=instructabl09-20\)](http://www.amazon.com/Herco%C2%AE-HE826-Precision-Screwdriver-Set/dp/B000CCUFT2/?tag=instructabl09-20)

(x1) [Scissor \(http://www.amazon.com/Westcott-13901-Sewing-Scissors-8-Inches/dp/B000P0LNRE/?tag=instructabl09-20\)](http://www.amazon.com/Westcott-13901-Sewing-Scissors-8-Inches/dp/B000P0LNRE/?tag=instructabl09-20)

(x1) [Razor blade \(http://www.amazon.com/Stanley-10-099-Classic-Retractable-Utility/dp/B00002X204/?tag=instructabl09-20\)](http://www.amazon.com/Stanley-10-099-Classic-Retractable-Utility/dp/B00002X204/?tag=instructabl09-20)

(x1) [Hacksaw \(http://www.amazon.com/XtremepowerUS-Adjustable-Hacksaw-Cutting-Bi-Metal/dp/B00OKXVGR4/?tag=instructabl09-20\)](http://www.amazon.com/XtremepowerUS-Adjustable-Hacksaw-Cutting-Bi-Metal/dp/B00OKXVGR4/?tag=instructabl09-20)

(x1) [Diagonal cutting pliers \(http://www.amazon.com/Channellock-338-8-Inch-Diagonal-Cutting/dp/B00004SBDD/?tag=instructabl09-20\)](http://www.amazon.com/Channellock-338-8-Inch-Diagonal-Cutting/dp/B00004SBDD/?tag=instructabl09-20)

(x1) [Multipurpose tools \(http://www.amazon.com/Leatherman-830039-Multitool-Leather-Combination/dp/B0002H49BC/?tag=instructabl09-20\)](http://www.amazon.com/Leatherman-830039-Multitool-Leather-Combination/dp/B0002H49BC/?tag=instructabl09-20)

(x1) [Adjustable wrench \(http://www.amazon.com/GRIP-87042-Adjustable-Wrench-4-Piece/dp/B000BN39MC/?tag=instructabl09-20\)](http://www.amazon.com/GRIP-87042-Adjustable-Wrench-4-Piece/dp/B000BN39MC/?tag=instructabl09-20)

(x1) [C-clamps \(http://www.amazon.com/TEKTON-91809-Heavy-Duty-C-Clamp-3-Piece/dp/B00BRL59HK/?tag=instructabl09-20\)](http://www.amazon.com/TEKTON-91809-Heavy-Duty-C-Clamp-3-Piece/dp/B00BRL59HK/?tag=instructabl09-20)

(x1) [Zip ties \(http://www.amazon.com/Cable-Matters-Combo-Self-Locking-12-Inch/dp/B00L2LGMO4/?tag=instructabl09-20\)](http://www.amazon.com/Cable-Matters-Combo-Self-Locking-12-Inch/dp/B00L2LGMO4/?tag=instructabl09-20)

(x1) [Hot glue guns \(https://www.amazon.com/Surebonder-DT-200KIT-Mini-Dual-Temperature/dp/B001AQRLM8/\)](https://www.amazon.com/Surebonder-DT-200KIT-Mini-Dual-Temperature/dp/B001AQRLM8/)

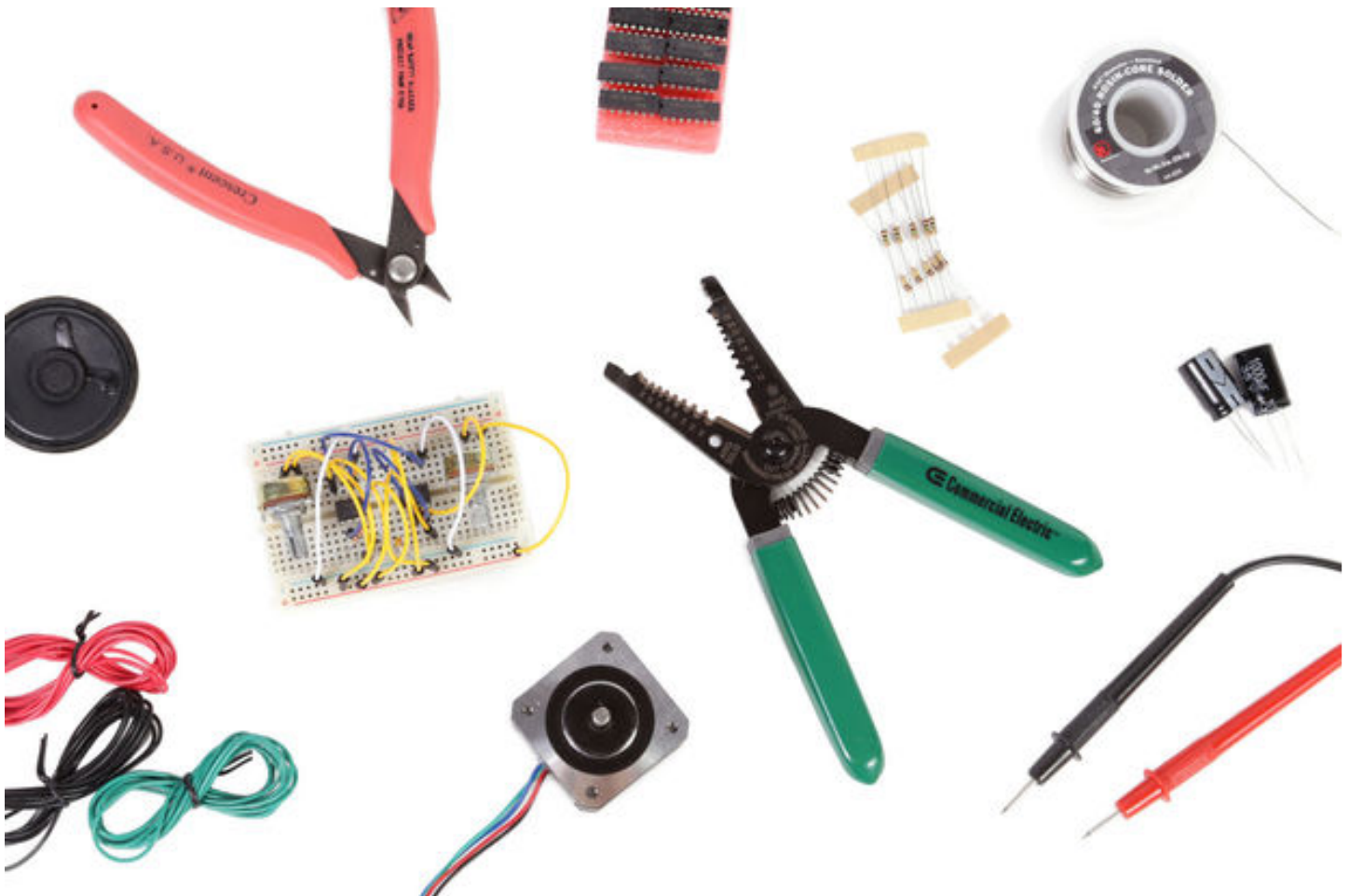
(x1) [Rulers \(http://www.amazon.com/Starrett-ASE-48-Anodized-Aluminum-Straight/dp/B002BXPUKO/?tag=instructabl09-20\)](http://www.amazon.com/Starrett-ASE-48-Anodized-Aluminum-Straight/dp/B002BXPUKO/?tag=instructabl09-20)

(x1) [Permanent marker \(http://www.amazon.com/Sharpie-Permanent-Markers-36-Pack-35010/dp/B001ELJOOM/?tag=instructabl09-20\)](http://www.amazon.com/Sharpie-Permanent-Markers-36-Pack-35010/dp/B001ELJOOM/?tag=instructabl09-20)

(x1) [Hammer \(http://www.amazon.com/Estwing-E3-16S-Straight-Hammer-Reduction/dp/B0000224VG/?tag=instructabl09-20\)](http://www.amazon.com/Estwing-E3-16S-Straight-Hammer-Reduction/dp/B0000224VG/?tag=instructabl09-20)

(x1) [Soldering iron kit \(https://www.amazon.com/gp/product/B01M0DLDCG/\)](https://www.amazon.com/gp/product/B01M0DLDCG/)

(x1) Exhaust fan (<https://www.amazon.com/Weller-WSA350-Bench-Smoke-Absorber/dp/B000EM74SK/>)



Basic Components Kit

To begin, it is advisable to get a [basic electronics kit \(https://www.amazon.com/Elenco-Component-Assorted-Electronic-Components/dp/B0002HBQHC/\)](https://www.amazon.com/Elenco-Component-Assorted-Electronic-Components/dp/B0002HBQHC/) which includes resistors, capacitors, diodes, and LEDs (amongst other things). There are no hobby kits which are perfect. They will always be missing that one last thing you need. However, the one I linked to includes a little bit of everything, and is a good place to start. You will be using lots of different components throughout the course, and having this on hand will come in useful for experimenting and trying things out.

You should also pick up [solid \(https://www.amazon.com/Electronix-Express--Hook-Wire-Solid/dp/B00B4ZRPEY/\)](https://www.amazon.com/Electronix-Express--Hook-Wire-Solid/dp/B00B4ZRPEY/) and [stranded \(https://www.amazon.com/Electronix-Express-Hook-Stranded-Gauge/dp/B00B4ZQ3L0/\)](https://www.amazon.com/Electronix-Express-Hook-Stranded-Gauge/dp/B00B4ZQ3L0/) wire, a few [breadboards \(https://www.amazon.com/Frentaly@-Solderless-BreadBoard-tie-points-power/dp/B01258UZMC/\)](https://www.amazon.com/Frentaly@-Solderless-BreadBoard-tie-points-power/dp/B01258UZMC/), a [multimeter \(https://www.amazon.com/Flexzion-Multimeter-Automotive-Electrical-Components/dp/B017QR594G/\)](https://www.amazon.com/Flexzion-Multimeter-Automotive-Electrical-Components/dp/B017QR594G/), and a [test amp \(https://www.radioshack.com/products/radioshack-mini-audio-amplifier?variant=5717614213\)](https://www.radioshack.com/products/radioshack-mini-audio-amplifier?variant=5717614213), all of which, I am about to explain further.



Wire

One of the most important things to have in your arsenal is wire.

You may be wondering what there is to say about wire? Well - a lot!

◦

In electronics, the wire we will be dealing with is insulated. This means that there is a metal core inside of a rubber or plastic sheath. This allows electricity to flow, but prevents the wires from shorting if they were to touch (because they are insulated).

◦

There are two types of metal cores we will be dealing with in this course.

Solid core wire has a single piece of metal inside the insulation. This type of wire is good for electronic circuit boards or connecting components together on a breadboard because it can easily plug into the board's sockets. Solid wire keeps its shape when bent, but it's also more prone to break if flexed too often.

Stranded core wire consists of thin metal strands bunched together. This wire is better for connecting to components which are handled a lot or move around (such as connecting to motors on a robot arm). This type of wire does not easily plug into a microcontroller's sockets, as the strands spread apart and fray, making it annoying for prototyping. However, it is very flexible and can be bent a lot without snapping.

◦

The thickness of wire is measured in gauges. The thicker the gauge, and the smaller the number rating, the more current it can handle. In America, gauge is measured in AWG.

We will largely be dealing with wire in the 20 AWG to 22 AWG range.

◦

Even though all wire essentially works the same regardless of color, there is a generally agreed upon color-coding system for wire when dealing with DC electronics.

Red indicates a power wire.

Black indicates a ground wire.

Green (or any color not red or black) indicates a signal wire.

While you are probably thinking we've exhausted all that there is say about wire, you would be wrong. However, there is still a lot more ground to cover and we should probably continue this wire discussion another day.

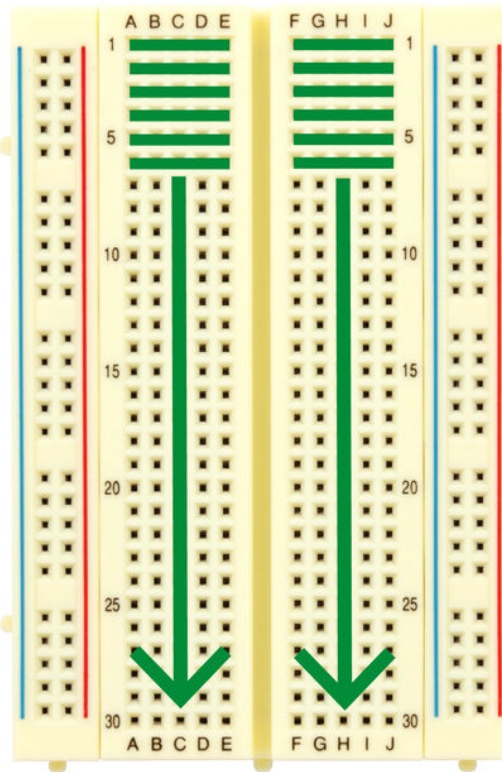


Breadboard

When you need to quickly and temporarily prototype a circuit, you will be using a breadboard.

Breadboards are meant to make quick non-permanent connections between electronic components. They are covered in tiny socket holes which are connected in rows. The board itself is broken into four sections. There are two inner sections full of short horizontal rows, and two outer sections with longer vertical rows.

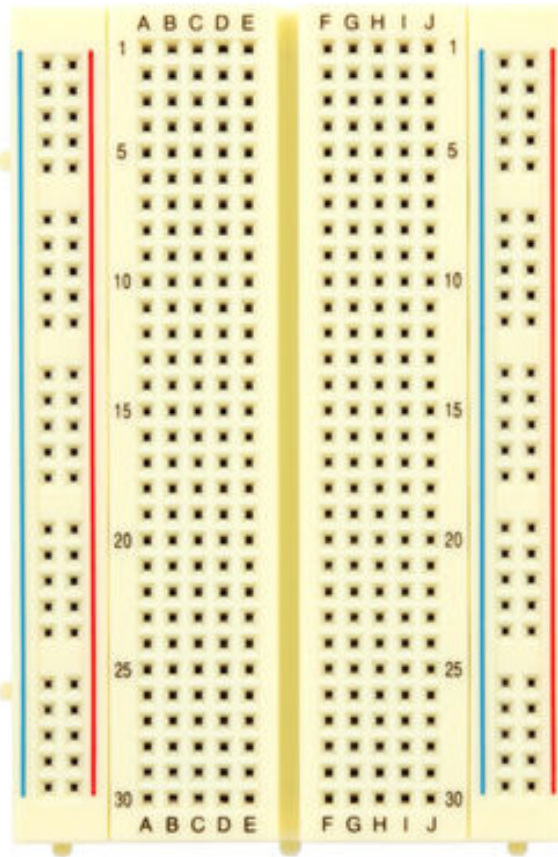
The inner sections are typically used for connecting components, and the outer sections are typically used as power bus lines. In other words, you can connect a battery to one of the outer lines and then power components on the inner section by connecting a wire to this section.



In the above graphic you can visually get a sense of how the rows on breadboards are electrically connected. The two inner sections have short horizontal rows repeated down the board. The two outer sections each have two long vertical rows. These are marked in red and blue and are meant to signify a row for power (red) and a row for ground (blue). Not all breadboards are marked with lines like this, but they are all laid out the same way.

To use a breadboard to prototype circuits, you simply insert components or wire into the appropriate sockets to connect them together.

On account of their ease of use for circuit building, it is best to have 2 or 3 breadboards on hand.



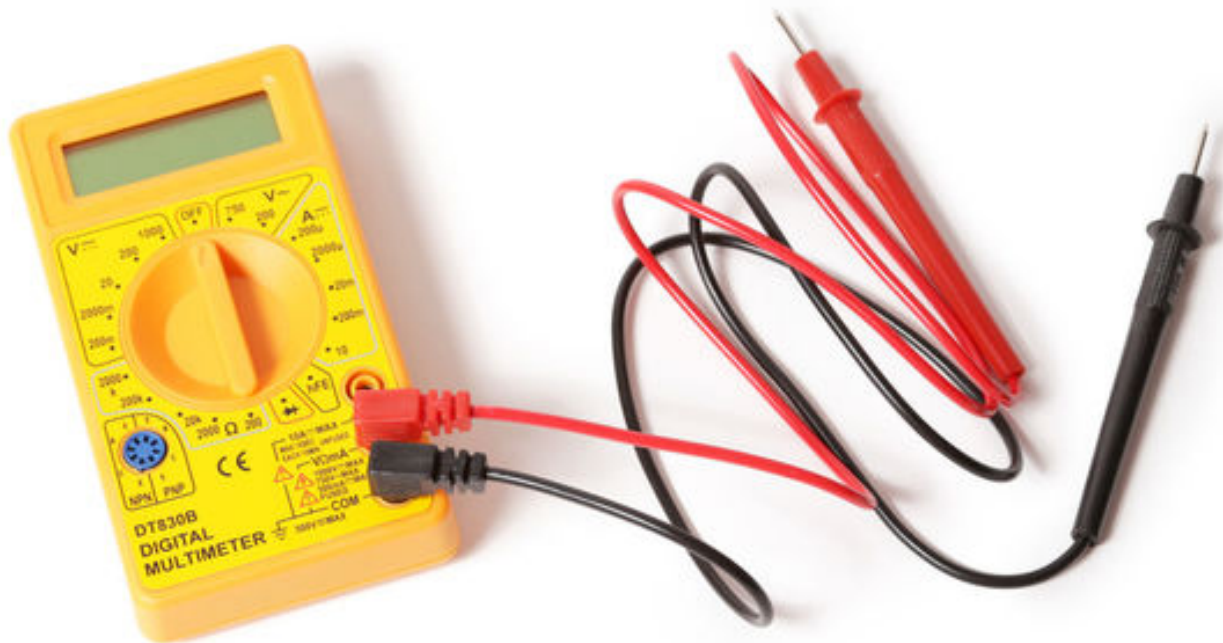
Multimeter

A multimeter is a tool used for a wide range of electronics related measurements. Or you could say they could meter multiple measurements... multi-meter... painfully obvious - right?

A typical multimeter will measure voltage, current, resistance, and continuity. More advanced multimeters will also measure a host of other things that are not relevant to get into at this junction.

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To set up the multimeter, plug the black probe into the ground / common port. Plug the red probe into the voltage terminal. It's now all set up.



Test Amp

A test amp is particularly useful when experimenting with components and producing an audible output. As you begin to learn electronics, you will be amazed how many circuits can make noise, and how frequently this device will prove useful.

The reason this particular mini amplifier comes in so useful is that it both has an input for audio, and an amplified output for driving external speakers.



Electronics Toolbox

As mentioned, working with electronics requires its own unique set of tools. We have already discussed a few, but here are a few more you will want to add to your tool box.

You will want both a wire cutter (<http://www.amazon.com/Tools-VISE-GRIP-Stripper-Crimper-2078309/dp/B000JNNWQ2/?tag=instructabl09-20>) and a pair of mini diagonal cutting pliers or "snippers" (<http://www.amazon.com/Hakko-CHP-170-Stand-off-Construction-21-Degree/dp/B017ODDPNO/?tag=instructabl09-20>). The wire cutter used for cutting and stripping insulation off of wires. The snippers are used for trimming away excess wire leads after you solder. When you are doing this for a while and start to get the hang of it, you can use snippers for everything (in place of the wire strippers).

Another indispensable set of tools is heat gun (<http://www.amazon.com/Kawasaki-840015-Black-10-Piece-Heat/dp/B000H4I67I/?tag=instructabl09-20>) and shrink tube (<http://www.amazon.com/SummitLink-Assorted-Shrink-Colors-Tubing/dp/B00LVFDLUO/?tag=instructabl09-20>). These are used for insulating soldered wire connections and small components. You might be thinking that electrical tape can do the same thing and is much cheaper. Just - NO! Get that horrible thought out of your head. Electrical tape is dumb and unreliable. If you want your circuits to break in mysterious ways, use electrical tape. If you want to make working circuits, then you should purchase shrink tube in a variety of colors and sizes. Shrink tube is exponentially more reliable and aesthetically pleasing. In terms of a heat gun, they all basically work the same. Just get something that makes you look cool. For instance, this black and green Kawasaki one is fairly rad.

Jumper cables (<http://www.amazon.com/SE-TL10-10-Piece-Alligator-Clips/dp/B0002KRABU/?tag=instructabl09-20>) (or test leads) are used for connecting wires together without soldering and are important for prototyping. They have insulated alligator clips on both ends which allow you to easily grab onto most electrical contacts. It is important to have these lying about to easily test things before making more permanent connections. Get a set of about a dozen-or-so to start.

Battery holders are used to power your projects. Typically, when one is required it is specified in the list of materials. However, in some of the lessons we use them for testing and experimenting. That said, it is recommended that you pick up a few extra 3 X AA and 4 X AA battery holders.

Extended Toolbox



Go get a drill (<http://www.amazon.com/DEWALT-DC970K-2-18-Volt-Compact-Driver/dp/B002RLR0EY/?tag=instructabl09-20>). This can be a cordless or corded drill. It does not matter. Cordless drills are more convenient in some ways, but corded drills are cheaper and just as (or sometimes more) effective. Either will do the job. We are only going to be drilling through wood, plastic parts and some soft metal like aluminum.

It is not important to get anything too fancy. Just about any drill will do for the purposes of this class. Albeit, it couldn't hurt to spend a little extra dough if you plan on continuing building things after this class. Nevertheless, the most important part is to find something aesthetically pleasing. It is always important to look good while making things.

Get a set of standard sized multipurpose drill bits (<http://www.amazon.com/DEWALT-DW1354-14-Piece-Titanium-Drill/dp/B0045PQ762/?tag=instructabl09-20>). If this is your first time doing something like this, any old set will do. Don't spend a lot of money. You will likely destroy them and need to buy another set at some point anyhow. As you start to figure out what you are doing, then you can invest in the fancy expensive drill bits.



Even though you can technically screw things very tight with your power drill, sometimes you just want to screw things the old fashioned way. It's good to have a range of screwdrivers (<http://www.amazon.com/Craftsman-Screwdriver-Phillips-Slotted-Made/dp/B006YVTAFU/?tag=instructabl09-20>) in your arsenal. While I am not going to dive too deep into this, I will say that you should get a set of mini screwdrivers (<http://www.amazon.com/Herco%C2%AE-HE826-Precision-Screwdriver-Set/dp/B000CCUFT2/?tag=instructabl09-20>). These will come in particularly handy when working with electronics.



You will also need a pair of scissors (<http://www.amazon.com/Westcott-13901-Sewing-Scissors-8-Inches/dp/B000P0LNRE/?tag=instructabl09-20>). You should already have one of these lying about, and learned how to use them if you ever attended Kindergarten. So... Moving on...

◦

The other razor sharp tool you should have is a razor blade (<http://www.amazon.com/Stanley-10-099-Classic-Retractable-Utility/dp/B00002X204/?tag=instructabl09-20>) or craft knife. It is recommended you get something with a nice safe handle like a box cutter.

◦

When a razor blade just won't cut it, you can be sure a hacksaw (<http://www.amazon.com/XtremepowerUS-Adjustable-Hacksaw-Cutting-Bi-Metal/dp/B00OKXVGR4/?tag=instructabl09-20>) will. With enough patience you can cut through most anything with a hacksaw. This hand saw will be extremely handy to have around.

◦

Diagonal cutting pliers (<http://www.amazon.com/Channellock-338-8-Inch-Diagonal-Cutting/dp/B00004SBDD/?tag=instructabl09-20>), or put simply, diagonal cutters, are another extremely useful tool to have at your disposal. It's recommended you get a good solid pair of these, and not one of those discount ones. These will be used a lot to cut through plastic, and thin metal parts. This tool is a bit like scissors for solid objects. No - it is exactly like scissors for solid objects.



Speaking of pliers, you are probably more familiar with the kind that don't cut. You should get at least one pair of general use snub nose pliers. Pictured here is a multipurpose tool centered around a pair of pliers. If you have the income at your disposal, you may as well invest in a nice [multipurpose tool](http://www.amazon.com/Leatherman-830039-Multitool-Leather-Combination/dp/B0002H49BC/?tag=instructabl09-20) (<http://www.amazon.com/Leatherman-830039-Multitool-Leather-Combination/dp/B0002H49BC/?tag=instructabl09-20>). The added functionality always comes in handy and it will make you seem more legit to have one of these in your arsenal. Again, I would like to reiterate the importance of looking like you really know what you are doing.

Adjustable wrenches (<http://www.amazon.com/GRIP-87042-Adjustable-Wrench-4-Piece/dp/B000BN39MC/?tag=instructabl09-20>) come in a wide range of sizes. You are likely not building anything too extreme and will be fine with just a small one (or two). These are largely used for tightening and loosening nuts and bolts, and you may be able to get away without it for this class.

Having a couple small C-clamps (<http://www.amazon.com/TEKTON-91809-Heavy-Duty-C-Clamp-3-Piece/dp/B00BRL59HK/?tag=instructabl09-20>) around can't hurt. In fact, they are meant to hold things down to prevent injury. It's good to have a few of these about for holding things when drilling or smashing things. Using your hand for this purpose can end in disaster.

Zip ties (<http://www.amazon.com/Cable-Matters-Combo-Self-Locking-12-Inch/dp/B00L2LGMO4/?tag=instructabl09-20>) are the greatest. They are quick, strong, reliable, and dirt cheap. If you make a mistake, they are exceptionally easy to undo. Building things using zip ties lets you iterate quickly and try out different solutions. You can chain them together, or trim them down. They can wrap around strange angles and be used to solve all kinds of problems. If you have not already deduced by my effusive praise, they will be the fastener of choice in this class. It is important that you get a lot of zip ties in a variety of sizes.

Less functional than zip ties, hot glue guns (<https://www.amazon.com/Surebonder-DT-200KIT-Mini-Dual-Temperature/dp/B001AQRLM8/>) are a quick and dirty way to attach things together. These come in particularly handy when you need to quickly hold an electronic component in place inside of an enclosure and don't feel like figuring out a way to mechanically fasten it.



Aside from making great construction material, it is very helpful to have a few rulers (<http://www.amazon.com/Starrett-ASE-48-Anodized-Aluminum-Straight/dp/B002BXPUKO/?tag=instructabl09-20>) around. As they say, 'measure twice - cut once.'

And, of course, if you are going to be employing rulers in your electronics activities, you got to have some permanent markers (<http://www.amazon.com/Sharpie-Permanent-Markers-36-Pack-35010/dp/B001ELJOOM/?tag=instructabl09-20>) to go along with them. We will be making a lot of cut and drill marks, and your marker will get a lot of mileage.



When all else fails, smash things.

We are only using one smashing tool - the trusty hammer (<http://www.amazon.com/Estwing-E3-16S-Straight-Hammer-Reduction/dp/B0000224VG/?tag=instructabl09-20>).

On occasion we will need to smash things. It is unavoidable.

Prepare Yourself

The parts list for each lesson is listed in the second step. I am not going to provide an overall shopping list for the class because some lessons give you a choice of projects to make, and it is best you review them before making any purchasing decisions.

Nevertheless, you should now have more than enough items on your shopping list to get going.

Take a deep breath and do whatever you need to do to relax - listen to music, light a candle, eat a doughnut - and get ready to start soldering.